NON-TECHNICAL REPORT: DEVELOPING CURIOSITY AND ANALYTIC INQUIRY

Project: BNPL Credit Risk Transformation

Subject: Strategic Curiosity & Question Formulation

Date: December 2025

1. Developing Curiosity and the Ability to Ask Good Questions

In the context of the Bati Bank BNPL project, curiosity is not just an interest in data; it is the engine that drives risk mitigation. When dealing with alternative data (where traditional credit scores are missing), a Data Scientist must look beyond the surface level of "who spent what."

True curiosity involves questioning the **intent** and **stability** of a customer. We moved from simply observing transactions to asking: *"Does this behavior indicate a sustainable lifestyle or financial distress?"* Developing this mindset allowed us to create a "Proxy Variable" that effectively mirrors real-world credit risk. By asking "Why?" at every stage of the EDA, we identified that transaction timing is often more telling than the transaction amount.

2. Curiosity Demonstrated: The Inquiry Process

Throughout this project, my analytical process was guided by exploring the "Unknowns." To build the improved fraud and risk detection model, I had to investigate several dimensions of the dataset:

- **The "Zero-Default" Problem:** How do we define a "bad" customer if no one has defaulted yet?
- **Temporal Patterns:** Are customers who shop at night riskier than those who shop during business hours?
- **Product-Risk Correlation:** Is a person buying "Airtime" more stable than someone using "Financial Services" transfers?

This process of constant questioning ensured that our final Machine Learning model was not just mathematically accurate, but business-relevant.

3. Top 5 Strategic Questions: Final Selection

After evaluating dozens of potential inquiries, the following five questions were selected as the most critical for Bati Bank's long-term BNPL success. These questions are sorted by their **strategic impact** on risk and profitability.

Rank	Strategic Question	Rationale for Selection
01	**What is the "Volatility Threshold" for a user's transaction frequency?**	Identifying the exact point where a drop in frequency predicts a future default allows for proactive risk intervention.
02	**How does the 'Category' of spend influence the probability of repayment?**	Not all transactions are equal. Understanding which services (e.g., utility bills vs. luxury airtime) correlate with stability is key to credit limit setting.
03	**Can 'Recency' serve as a real-time proxy for sudden financial shocks?**	If a high-frequency user suddenly stops for 10 days, the model must flag this as a potential life event that increases credit risk.
04	**What is the optimal 'First-Loan' amount to minimize initial loss?**	Curiosity about the "Entry Point" helps the bank determine the safest starting credit limit for a user with no history.
05	**Is there a seasonal pattern to high-risk transactions?**	Understanding if risk increases during holidays or month-ends allows the bank to tighten or loosen credit filters dynamically.

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4. Writing Quality and Presentation Summary

This non-technical submission serves as a bridge between data engineering and executive decision-making.

- **Clarity:** The language used avoids jargon to ensure stakeholders in Finance and Marketing can act on the insights.

- **Strategic Value:** The questions selected focus on **Profitability** and **Loss Prevention**, the two highest priorities for Bati Bank.
- **Presentation:** The use of structured tables and clearly defined sections ensures that the "Curiosity" aspect of the project is seen as a formal business asset rather than a secondary thought.